INT-016 Purification Process Description

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Function/ Department	Responsibilities
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Prepared by: Purification Process Sciences	Responsible for identifying and defining all project specific parameters and values for for transfer to manufacturing • Identify required parameters and define values for targets and ranges for all project specific parameters including buffer recipes based on development experience to ensure appropriate product quality and process performance • Compile process updates and justifications for variations relative to platform process (ED000) or prior process description edition in the Process Changes Section
Approved by: Formulation Sciences	Responsible for reviewing all formulation specific parameters and verifying that they are accurate and in alignment with current FS strategy Sections: 1.1, 12-15 • Define final formulation, container, and room temperature hold time • Verify that buffer recipes used during formulation are accurate • Verify that densities are sufficiently accurate
Approved by: Purification Process Sciences	Responsible for reviewing project specific parameters and confirming that they are in alignment with current PPS strategy (GUID-0024008) • Confirm that variations from platform process are captured in Process Changes and justified
Approved by: Purification Process Sciences Technology Transfer	Responsible for reviewing pertinent information/ parameters for transfer to manufacturing • Parameter targets and ranges are in alignment with general PPS knowledge and expectations and facility capabilities • Maintain consistency across projects and process descriptions

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1 INTRODUCTION

1.1 Purpose

This report describes the platform purification process and process parameters for the production of INT-016. The overall process (upstream/downstream) is designated “Process 2”.

Table 1-1 General Properties: mAb Intermediate

Property	Description
Target	B7H4
Protein classification	monoclonal antibody (mAb) IgG1
Isoelectric point (pI) range (experimental)	9.35
Extinction coefficient	1.58 (Theoretical)
Expected titer range (g/L) 90% Prediction Interval	8.39 (6.92 - 9.86)
Overall expected yield from recovery through mAb Intermediate (%)	60-70%
mAb intermediate concentration (g/L)	50
mAb intermediate buffer	20 mM histidine/histidine-HCl, 240 mM sucrose, pH 6.0
mAb intermediate presentation	Stored frozen at ≤-40°C in CryoVaults Or Stored frozen at ≤ -65°C in Cryovaults

1.2 Assumptions

- This report does not consider scale or facility constraints.
- Unless otherwise noted, procedures pre/post common unit operations should be followed for filters and unit operations per GUID-0024075.
- Unless otherwise noted, all conductivity values are compensated to a reference temperature of 25°C using a temperature coefficient (θ) of 2.0% per °C.
- The target volumes are expected to be sufficient. Overage to account for system hold up volume is site dependent and should be a separate phase and not added into the volume targets for the process.
- Unless otherwise noted, flow rate ranges are for steady state operation and do not consider pump ramping strategy.
- Unless otherwise noted, all unit operations are conducted at controlled room temperature (15-27°C).
- For facilities for which 1.0 N NaOH is not feasible, 0.5 N NaOH may be used as a substitute with “Conductivity (mS/cm @ 25°C), final” NOR: 84 – 113.
- Unless otherwise noted, all chromatography step phases are operated in down flow mode.
- Unless otherwise noted, all process parameters are inputs.

1.3 Abbreviations/Acronyms/Definition

Table 1-2 Abbreviations and Acronyms

Acronym	Definition
AEX	Anion exchange chromatography
AL	Alert limit
Bldg	Building
BR	Batch record
CEX	Cation exchange chromatography
CM	Conditioned media
Conc.	Concentration
CV	Column volume
DV	Diavolumes
DBC	Dynamic binding capacity
DCF	Density Correction Factor
DS	Drug Substance
DSC	Development Scale-Up Center (500-L scale)
EAB	Excipient addition buffer
ELISA	Enzyme-linked immunosorbent assay
EP	European Pharmacopeia
FB	Formulated bulk
FIO	For information only
FF	Fast Flow
FSS	Formulation Spiking Solution
FMC	Frederick Manufacturing Center
GPF	Gaithersburg Pilot Facility
GMP	Good Manufacturing Practices
HETP	Height equivalent to a theoretical plate
HIC	Hydrophobic interaction chromatography
HPLC	High-performance liquid chromatography
HPSEC	High-performance size-exclusion chromatography
HCP	Host cell protein
IgG1/2/4	Immunoglobulin G1/2/4
IEC	Ion exchange chromatography
JP	Japanese Pharmacopoeia
kDa	Kilodalton
LF	Load Filter

Acronym	Definition
LG	Linear gradient
LR	Lot release
LMH	Liters/meter^2-hour
MV	Membrane volume
MWCO	Molecular Weight Cut Off
NA	Not applicable
NC	Non-conformance
NF	National Formulary
NOR	Normal operating range
OD	Optical density
PCR	Polymerase chain reaction
PES	Polyethersulfone
PF	Product Filter
PFI	Pre-formulated intermediate
P/N	Part number
PP	Polypropelyne
psid; psig	pounds per square inch differential; pounds per square inch gauge
PS80	Polysorbate 80
Protein A	Protein A affinity chromatography
qs	Bring to final weight (quantum satis)
RPHPLC	Reverse phase high performance chromatography
S.F.	Safety factor
SS	Stainless steel
SUL	Scale-Up Lab
TBD	To be determined
TMP	Trans-membrane pressure
UB	Unformulated Bulk
UDS	Unformulated Drug Substance
USP	United States Pharmacopoeia
VC	Viral clearance
VF	Virus filtration
VPro	Viresolve Pro
WFI	Water for injection

1.4 Definitions of Terminologies for Parameter Ranges

1.4.1 Target

The target value is a preset value that a process parameter is aimed to by process controls. The controls include any combination of manual or automated, discrete or continuous, feed-forward or feed-back responses to operate the process near the target value. Where Targets are provided, the process should be controlled to meet the target value as closely as possible, not just controlling to the edge of the Alert Limit or Normal Operating Range.

1.4.2 Normal Operating Range (NOR)

Operating parameter ranges, also called Action Limits, which are established based on current understanding and process knowledge gained through development and process characterization. NORs are specified in manufacturing batch records and/or other GMP documentation (such as automation recipes) to control the process. A value exceeding a NOR triggers a non-conformance investigation and is documented and evaluated according to the site's Quality System.

2 BILL OF MATERIALS

Table 2-1 Bill of Materials

Type	Component Name	Minimum Grade	MEDI P/N	Justification/Notes
Chromatography media	MabSelect SuRe (Protein A)	NA	4101797 4102160 6001979 6002000	5 L (ethanol) 10 L (ethanol) 5 L (benzyl) 10 L (benzyl)
	Capto Q (AEX)	NA	6001609 6001621	NA
	Poros 50 HS (CEX)	NA	40003 4102211	NA
Buffer and solution components	Acetic acid, glacial	USP, EP	4101341 6000325	NA
	Benzyl alcohol	NF, EP, JP	4102670	NA
	L-histidine	USP, EP, JP	6001825	NA
	L-histidine HCl, monohydrate	EP, JP	4101011	NA
	Sodium acetate, trihydrate, crystal	USP, EP, JP	4101342 6000326	NA
	Sodium chloride	USP, EP	6000324	NA
	10N Sodium hydroxide	NA	4102099 4102229	Biotech Reagent
	Sodium hydroxide, pellets	EP, NF	4101097	NA
	Sucrose (low endotoxin)	EP, NF, JP	4102624	NA
	Tris base (tromethamine/trometamol)	USP, EP	4101399	NA
	Tris hydrochloride	NA	4100924	Biotech Reagent
Purification filters	Viresolve Pro Magnus Device	NA	4102501 4102503	0.51 m^2 1.53 m^2
	Viresolve Pro Magnus Shield	NA	4102530 4102531	0.51 m^2 1.53 m^2
	Pellicon 3 Ultracel C Screen ultrafiltration cassettes, 30kDa MWCO	NA	6001576 4102051 4102050	0.11 m^2 0.57 m^2 1.14 m^2
	Millistak+ HC depth Pod filters-X0HC	NA	4101860	1.1 m^2
	Durapore Opticap, 0.22 µm	NA	NA	Sterile Grade 0.2 µm
	Millipore Express SHC, 0.5/0.2 µm	NA	NA	Generic 0.2 µm
	Pall Fluorodyne EX EDF, 0.2 µm	NA	NA	Generic 0.2 µm
Miscellaneous components	Meissner CryoVaults	NA	6003782 6003895	20 L

3 IN-PROCESS HOLD TIMES

Process intermediate stability studies were performed to support pooling of chromatography cycles and holding of process intermediates.

Hold times begin when product is collected and end when the next step is loaded.

Unless otherwise noted, room temperature hold times are inclusive to refrigerated temperature hold times but not vice versa. For example, a product that was held for 48 hours at room temperature prior to being stored 24 hours refrigerated would have a total hold time of 72 hours at refrigerated temperature but only 48 hours at room temperature.

Table 3-1Process Intermediate Hold Times

Process Intermediate	Room Temperature Hold Time - Max (days, 15-25 °C)	Refrigerated Temperature Hold Time - Max (days, 2-8 °C)	Justification/Notes
Conditioned medium	1	7	NA
Protein A Product	2	7	Hold times for Protein A and VI product is cumulatively 3 days max at 15-25 °C
Virus Inactivation product Pre-filter	1	7	Virus inactivation product hold times are cumulative for pre and post filter. 2 options are provided to facilitate facility fit.
Virus Inactivation Product, Post filter	1	7	
AEX Product	3	7	NA
CEX Load Adjustment Product	3	7	NA
CEX Product	5	14	NA
CEX adjusted product	5	14	Leverages CEX Product Data
Virus Filtration Product	5	14	Leverages CEX Product Data
UFDF Recovery Product	5	14	Leverages PFI Product data
Pre-formulated Intermediate	5	14	NA
Formulated Product (pre-final filtration)	3	7	Leverages final mAb intermediate data

4 LOAD AND PRODUCT FILTRATION

Table 4-1Filtration Parameters

Unit Operation	Position	Membrane Type	In-line	Volumetric Load NOR-Max (L/m2)	Justification/Notes
Protein A Chromatography (LF1 →LF2→ Unit Op → PF)	Load filter 1	Millistak+ HC POD filters - X0HC	Load-Equilibration 2	558	Max challenge at DSC is 558 L/m^2 for lot MS01281-137 with minimal pressure increase observed
	Load Filter 2	Express SHC, 0.5/0.2 µm	All Column Steps In-line	3000	Max challenge at DSC is 788 L/m^2 for lot MS00684-89 with minimal pressure increase observed
	Product Filter[a]	Express SHC, 0.5/0.2 µm	Collection	3000	Max challenge at GPF is 720 L/m^2, minimal pressure increase observed
Virus Inactivation (Unit Op → PF 1→ PF 2)	Product Filter 1	Millistak+ HC POD filters - X0HC	Off-line[b]	320	Max challenge at DSC 300 L/m^2 for lot MS00684-89 with minimal pressure increase observed
	Product Filter 2	Express SHC, 0.5/0.2 µm	Off-line	3000	Max challenge at GPF is 461 L/m^2, minimal pressure increase observed
AEX (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Column Steps In-line	3000	Max challenge at GPF is 507 L/m^2 , minimal pressure increase observed
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3000	Max challenge at DSC is 1110 L/m^2 for lot MS00684-89 with minimal pressure increase observed

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Table 4-1Filtration Parameters

Unit Operation	Position	Membrane Type	In-line	Volumetric Load NOR-Max (L/m2)	Justification/Notes
CEX (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Column Steps In-line	3000	Max challenge at GPF is 592 L/m^2, minimal pressure increase observed
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3000	Max challenge at DSC is 738 L/m^2, for lot MS00684-63 . Pressure data not available.
Virus Filtration (Unit Op → PF)	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3000	Max challenge at DSC 1693 L/m^2 for lot MS00684-63 with minimal pressure increase observed
Concentration and Diafiltration (Unit Op → PF)	Product Filter	Express SHC, 0.5/0.2 µm	Off-line	483	Max challenge at DSC is 483 L/m^2 for lot MS00684-89 with minimal pressure increase observed Recommend smallest device within range
mAb Intermediate Formulation and Bulk Filtration (Unit Op → Bulk Filter)	Bulk Filtration	Durapore Opticap, 0.2 µm	Off-line	641	Max challenge at DSC is 641 L/m^2 for lot MS00684-89 with minimal pressure increase observed Recommend smallest device within range

Millipore SHC filters may be interchanges with Pall Fluorodyne EX EDF filters of the same device size. This assumes the additional filter area of the Pall device compensates for any difference in maximum volumetric load.

[a] Product filtration optional, based on facility fit.

[b] Filtration can occur inline with AEX chromatograpghy load, based on facility fit.

5 PROTEIN A CHROMATOGRAPHY

Table 5-1 Protein A Chromatography: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	MabSelect Sure		Prep per appropriate SOP
Equilibration 1 *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
	pH, effluent, (endpoint check)[a]	NA	7.0 – 7.8	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	NA	Run and record.
Load *Conditioned Medium*	Linear velocity (cm/hr)	300	≤ 330	NA
	Protein load (g/L resin)	NA	≤ 40	NOR-Max is the capacity
Equilibration 2 *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	Add depth filter hold-up volume to the target
Wash *50 mM tris,* *0.5 M sodium chloride,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
Equilibration 3 *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA

Table 5-1 Protein A Chromatography: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
Elution *25 mM sodium acetate, pH 3.6*	Linear velocity (cm/hr)	150	≤ 330	NA
	Product collection start (OD_{280})	0.5	≥ 0.3	Volume before peak = ~ 0.6 CV
	Product collection end (OD_{280})	0.5	≥ 0.3	Product volume = ~ 1.2-1.6 CV Step yield = ~ 95%
Acid Strip *100 mM acetic acid*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
Equilibration 4 *50 mM tris, pH 7.4* or *WFI*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	1.0	≥ 0.5	NA
NaOH Strip *0.1 N sodium hydroxide*	Linear velocity (cm/hr)	300	≤ 330	
	Volume (CV)	3.0	≥ 2.5	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	19-27	NA
End of Cycle				
Sanitization Hold	Contact time (minutes)	35	30 -120	NA
Storage *2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
	pH, effluent, (endpoint check) [a]	NA	4.8-5.2	NA

[a] Process Output

6 VIRUS INACTIVATION (LOW PH TREATMENT)

Table 6-1 Virus Inactivation: Process Parameters

Step	Process Parameter (units)	Target	NOR/Action Limit	Justification/Notes
Acid Titration *1 M acetic acid*	pH, (endpoint check) [a]	3.5	3.4 – 3.6	~ 0.13 (kg acid / kg Product)
Incubation	Duration (minutes)	70	60 – 240	NOR-Min is based on viral clearance recommendation. NOR-Max is based on development stability data.
	Temperature (°C)	NA	18 – 25	Confirm temperature at start and end of incubation
	pH	NA	3.4 – 3.6	Confirm pH at start and end of incubation
Base Titration *1 M tris base*	pH, (endpoint check) [a]	7.4	7.2 – 7.6	~ 1.5 (kg base / kg acid) Step yield = ~ 100%

[a] Process output

7 AEX CHROMATOGRAPHY

Table 7-1 AEX Chromatography: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	Capto Q		Prep per appropriate SOP
Pre-Equilibration *50 mM tris,* *0.5 M sodium chloride,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
Equilibration *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
	pH, effluent, (endpoint check) [a]	NA	7.2-7.6	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	NA	Run and record.
Load *Virus Inactivation Product*	Linear velocity (cm/hr)	300	≤ 330	NA
	Protein load (g / L resin)	NA	≤ 100	NOR-Max is based on VC study.
	Product collection start (OD_{280})	0.5	≥ 0.3	Volume before peak = ~ 0.4 CV
Chase *50 mM tris,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Product collection end (OD_{280})	0.5	≥ 0.3	Product dilution = ~ 115% Step yield = ~ 96%

Table 7-1 AEX Chromatography: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
Salt Strip *50 mM tris,* *0.5 M sodium chloride,* *pH 7.4*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
NaOH Strip *1.0 N sodium hydroxide*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	164-211	NA
WFI Flush **(except last cycle)**	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
End of Cycle				
Sanitization Hold	Contact time (minutes)	35	15 — 120	NA
Pre-Storage Flush *WFI or 50mM tris,* *0.5M sodium chloride, pH 7.4*	Linear Velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	3.0	≥ 2.5	NA
Storage *2% benzyl alcohol, 100mM sodium acetate, pH 5.0*	Linear velocity (cm/hr)	300	≤ 330	NA
	Volume (CV)	5.0	≥ 2.5	NA
	pH, effluent, (endpoint check) [a]	NA	4.8-5.2	NA

[a] Process Output

8 AEX PRODUCT ADJUSTMENT

Table 8-1 AEX Product Adjustment: Process Parameters

Step	Process Parameter (units)	Target	NOR/Action Limit	Justification/Notes
pH Adjustment *1 M acetic acid*	pH	5.0	4.8 – 5.2	~ 0.06 (kg titrant / kg AEX Product)

9 CEX CHROMATOGRAPHY

Table 9-1 CEX Chromatography: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	Poros HS 50		Prep per appropriate SOP
Equilibration 1 *50 mM sodium acetate,* *50 mM Tris,* *pH 5.0*	Linear velocity (cm/hr)	300	≤ 330	Target 150 cm/hr linear velocity for Column bed height between 22- 25 cm
	Volume (CV)	3.0	≥ 2.5	NA
	pH, effluent, (endpoint check) [a]	NA	4.8-5.2	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	NA	Run and record.
Load *pH Adjusted AEX Product*	Linear velocity (cm/hr)	300	≤ 330	Target 150 cm/hr linear velocity for Column bed height between 22- 25 cm
	Protein load (g / L resin)	NA	≤ 40	NOR-Max is the capacity.
Equilibration 2 *50 mM sodium acetate,* *50 mM Tris,* *pH 5.0*	Linear velocity (cm/hr)	300	≤ 330	Target 150 cm/hr linear velocity for Column bed height between 22- 25 cm
	Volume (CV)	3.0	≥ 2.5	NA
Elution *50 mM sodium acetate,* *50 mM tris,* *200 mM sodium chloride* *pH 5.0*	Linear velocity (cm/hr)	150	≤ 330	NA
	Product collection start (OD_{280})	0.5	≥ 0.3	Volume before peak = ~ 0.8 CV
	Product collection end (OD_{280})	1.5	≥ 1.0	Product volume = ~ 1.5-3.8 CV[b] Step yield = ~ 94%

Table 9-1 CEX Chromatography: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
Salt Strip *50 mM acetate,* *1 M sodium chloride,* *pH 5.0*	Linear velocity (cm/hr)	300	≤ 330	Target 150 cm/hr linear velocity for Column bed height between 22- 25 cm
	Volume (CV)	3.0	≥ 2.5	NA
NaOH Strip *1.0 N sodium hydroxide*	Linear velocity (cm/hr)	300	≤ 330	Target 150 cm/hr linear velocity for Column bed height between 22- 25 cm
	Volume (CV)	3.0	≥ 2.5	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	164-211	NA
End of Cycle				
Sanitization Hold	Contact time (minutes)	35	15 — 120	NA
Storage *0.1 N sodium hydroxide*	Linear velocity (cm/hr)	300	≤ 330	Target 150 cm/hr linear velocity for Column bed height between 22- 25 cm
	Volume (CV)	3.0	≥ 2.5	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) [a]	NA	19-27	NA

[a] Process Output

[b] Elution is pH and salt sensitive, low end of pH and salt concentration could cause peak tailing

10 CEX PRODUCT ADJUSTMENT

Table 10-1CEX Product Adjustment: Process Parameters

Step	Process Parameter (units)	Target	NOR/Action Limit	Justification/ Notes
Concentration Adjustment *50 mM sodium acetate,* *50 mM tris,* *200 mM sodium chloride,* *pH 5.0*	Protein Concentration (g/L)	19	≤ 22	Adjustment only required if CEX Product concentration >22 g/L

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11 VIRUS FILTRATION

Table 11-1 Virus Filtration: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	Viresolve Pro+		Prep per appropriate SOP
	Feed pressure (psig)	NA	0-60	NA
Equilibration *50 mM sodium acetate,* *50 mM tris,* *200 mM sodium chloride,* *pH 5.0*	Differential pressure (psid)	30	0-60	
	Volumetric load (L/m^2)	75	≥ 65	NA
	pH, effluent, (endpoint check) [a]	NA	NA	Run and record.
Load *Adjusted CEX Chromatography Product*	Differential pressure (psid)	30	0-45	Based on process Acceptable Range maximum of 50 psid per report: *Justification of Upper Pressure Limit for VPro Virus Filter Operation*
	Total volumetric load (L/m^2 filter area)	NA	≤ 900	NOR-Max is based on VC study. Measured from the start of product load to the end of post-filtration flush (Equilibration 2)
	Flow decay (%) [a]	NA	≤ 90	100% × (Initial buffer flux-Final virus filtration flux) ÷ (Initial buffer flux)
	Product collection start (OD_{280})	0.5	NA	NA
Chase *50 mM sodium acetate,* *50 mM tris,* *200 mM sodium chloride, pH 5.0*	Differential pressure (psid)	30	0-45	Based on process Acceptable Range maximum of 50 psid per report: *Justification of Upper Pressure Limit for VPro Virus Filter Operation*

Step	Process Parameter (units)	Target	NOR	Justification/Notes
	Product collection end (OD_{280})	0.5	NA	Volume chase may be set if OD is not available. Step yield = ~100%

[a] Process Output

12 CONCENTRATION AND DIAFILTRATION

Table 12-1 Concentration and Diafiltration: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
General	Media	Pellicon 3 Ultracel C Screen 30kDa MWCO		Prep per appropriate SOP
	Feed pressure (psig)	NA	≤ 60	NA
Equilibration *20 mM histidine/histidine-HCl, pH 5.9*	pH, permeate, (endpoint check) [a]	NA	5.7 – 6.1	NA
Concentration 1 Control Method: Constant TMP and Feed Flow Control	Membrane protein load (g/m^2)	NA	≤ 1600	(Mass of protein)/(area of ultrafiltration membrane) NOR-Max is the tested capacity.
	TMP (psi)	20	≤ 35	NA
	Fed batch ratio	NA	NA	(Load volume)/(DCF tank volume)
	Feed flow (L/min/m^2)	5.0	3.0-7.0	May lower to maintain constant TMP.
	Protein concentration, endpoint check (g/L)	70	≤ 110	May be directly measured or calculated from permeate or retentate volume
Diafiltration *20 mM histidine/histidine-HCl, pH 5.9* Control Method:	TMP (psi)	20	≤ 35	NA
	Feed flow (L/min/m^2)	5.0	3.0-7.0	May lower to maintain constant TMP
	Number of diavolumes (DV)	6.0	≥ 5.5	(kg buffer)/(kg product)
	pH, permeate, (endpoint check) [a]	NA	NA	NA

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Table 12-1 Concentration and Diafiltration: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
Constant TMP and Feed Flow Control	pH, retentate, (endpoint check) [a]	NA	5.8 – 6.2	NA
Concentration 2 Control Method: Constant TMP and Feed Flow Control	TMP (psi)	20	≤ 35	NA
	Feed flow (L/min/m^2)	5.0	≤ 7.0	May lower to maintain constant TMP
	Protein concentration, endpoint check (g/L)	100	63.4 - 110	May be directly measured or calculated from permeate or retentate volume
Recirculation	Feed pressure (psig)	25	≤ 60	NA
	Feed flow (L/min/m^2)	NA	≤ 7	NA
	Recirculation time (min)	5	≤ 120	NA
Recovery Flush *20 mM histidine/histidine-HCl, pH 5.9* Recovery Method: Buffer flush of system hold up volume	Feed pressure (psi)	15	≤ 60	NA
	Feed flow (L/min/m^2)	NA	≤ 7	NA
	Protein concentration, endpoint check (g/L)	85.0	63.4-110	NA
Dilution to Pre-formulated Intermediate *20 mM histidine/histidine-HCl, pH 5.9*	Protein concentration, PFI (g/L)	66.7	63.4-70.0	NA
	pH, (endpoint check) [a]	NA	5.8 – 6.2	NA

[a] Process Output

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Table 12-2 Concentration and Diafiltration: Density Values

Buffer	Step	Protein Concentration (mg/mL)	Density (g/mL)	Notes
50 mM sodium acetate, 50 mM tris, 200 mM sodium chloride, pH 5.0	Conc. 1	0 – 10	1.011	Density = (0.000285 × protein concentration) + buffer density
	Conc. 1	10 – 20	1.014	NA
	Conc. 1	20 – 30	1.017	NA
	Conc. 1	30 – 40	1.020	NA
	Conc. 1	40 – 50	1.023	NA
	Conc. 1	50 – 60	1.026	NA
	Conc. 1	60 – 70	1.029	NA
20 mM histidine/histidine-HCl, pH 5.9	PFI	66.7	1.020	NA
	Diafiltration	70	1.021	NA
	Conc. 2	70 – 80	1.022	NA
	Conc. 2 (Recovery)	80 – 90 (85)	1.025	NA
	Conc. 2	90 – 100	1.028	NA
	Conc. 2, final	100	1.030	NA

13 FORMULATION

Table 13-1 Formulation: Process Parameters

Step	Process Parameter (units)	Target	NOR/Action Limit	Justification/Notes
EAB Spike (of PFI to FB) *EAB:* *20 mM histidine/histidine-HCl,* *960 mM sucrose,* *pH 6.0* (density 1.127 g/mL)	Ratio of EAB spike to PFI (kg/kg)	0.370	0.362 – 0.380	mAb Intermediate: 50 g/L INT-016, 20 mM histidine/histidine-HCl, 240 mM sucrose, pH 6.0 (experimental density 1.046 g/mL)
	Protein concentration, endpoint check (g/L) [a]	50.0	45.0 – 55.0	NA
	Osmolality (mOsm/kg) [a]	300	255 – 345	NA

[a] Process output . For the purpose of this document , these outputs must pass their acceptance criteria to ensure that product quality is maintained, and any result outside of those criteria triggers a deviation investigation.

14 BULK FILTRATION

Table 14-1 Bulk Filtration: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
	mAb intermediate container	CryoVaults		
Buffer Flush *EAB:* *20 mM histidine/histidine-HCl,* *960 mM sucrose,* *pH 6.0*	Buffer flush amount (L/m2)	NA	≥ 11	
Product Flush	Product flush amount (L/m2)	NA	≥ 1.9	
0.2µm Bulk Filtration	Fill volume per container (L)	NA	4.0 – 16.0	If multiple containers are required, target equivalent volume in each.
	mAb intermediate storage temperature (°C)	-40	- 45 – -35	mAb intermediate can be held 1 month at 2-8°C prior to freeze. mAb intermediate long term storage condition is ≤ -35°C. The antibody can be stored at a target of -40°C or -80°C depending on the available storage conditions per facility.
		-80	≤ -65	
Integrity Test *0.2 µm Filter*	Integrity Test Result (PASS/FAIL) [a]	PASS	PASS	Perform per appropriate SOP. Integrity test is an in-process control that must pass for the lot to be released

Table 14-1 Bulk Filtration: Process Parameters

Step	Process Parameter (units)	Target	NOR	Justification/Notes
	mAb intermediate container	CryoVaults		

[a] IPC: In-Process Controls (IPC) are defined in ICH Q7 as "Checks performed during production in order to monitor and, if appropriate, to adjust the process and/or to ensure that the intermediate or API conforms to its specifications. For the purposes of this document, IPCs are in-process tests or measurements which are not controlled directly, but which must pass their acceptance criteria to ensure that product quality is maintained, and any result outside those acceptance criteria triggers a deviation investigation.

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15 BUFFERS AND SOLUTIONS

Buffer and solution stability studies were performed by holding solutions and buffers at ambient temperature in stainless steel containers. The studies were designed to test chemical stability for longer than the expected maximum time for routine storage. These times may be extended by an appropriate study unless otherwise noted. The studies do not address microbial control, which should be defined by the facility. Assays used to measure stability included pH and conductivity. In some cases, the hold time is measured directly (direct). In other cases, the hold time is supported by the hold time of similar buffers if the proposed buffer falls within our current buffer matrix (matrix). Please see REP-0202700 and PRO-0188988 for reference.

Table 15-1Buffer and Solution Stability

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
50 mM tris, pH 7.4	PRO-0188988	8003734/ A-005378	35 (direct)	Protein A	Depth filter Equilibration
					Equilibration 1
					Equilibration 2
					Equilibration 3
					Equilibration 4
				Virus Inactivation	Depth filter Equilibration
					Depth filter Chase
				AEX	Equilibration 1
					Chase
					Equilibration 2
50 mM tris, 0.5 M sodium chloride, pH 7.4	PRO-0188988	8003737/ A-005380	35 (matrix)	Protein A	Wash
				AEX	Pre-Equilibration
					Salt Strip
					Pre-Storage

Table 15-1Buffer and Solution Stability

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
25 mM sodium acetate, pH 3.6	PRO-0188988	8002814/ A-005286	30 (matrix)	Protein A	Elution
100 mM acetic acid	PRO-0188988	A-005041/ A-005041	30 (direct)	Protein A	Acid Strip
WFI	NA	NA	NA	Protein A	Equilibration 4
				AEX	WFI flush
				AEX	Pre-storage flush
0.1 N sodium hydroxide	PRO-0188988	A0014-D/ A0014-D	7 (stedim bag) 90 (hyclone bag)	Protein A	NaOH Strip
				CEX	Storage
2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0	PRO-0188988	8002256/ A-005212	30 (direct)	Protein A	Storage
				AEX	Storage
1 M acetic acid	PRO-0188988	A-005168/ A-005168	77 (direct)	Virus Inactivation	Acid Titration
				AEX Product Adjustment	Adjustment
1 M tris base	PRO-0188988	A0101-D/ A0101-D	35 (matrix)	Virus Inactivation	Base Titration
50 mM sodium acetate, 50 mM Tris pH 5.0	PRO-0188988	8008534/ AZDoc0158 942	45	CEX	Equilibration 1
					Equilibration 2
50 mM sodium acetate, 50 mM Tris 200 mM sodium chloride, pH 5.0	PRO-0188988	8008535/ AZDoc0158 136	45	CEX	Elution
				CEX Product Adjustment	Adjustment
				Virus Filtration	Equilibration 1
					Equilibration 2

Table 15-1Buffer and Solution Stability

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
50 mM acetate, 1 M sodium chloride, pH 5.0	PRO-0188988	8006344	45	CEX	Salt Strip
1.0 N sodium hydroxide	PRO-0188988	A0191-D/ A0191-D	180 (direct)	AEX	NaOH Strip
				CEX	NaOH Strip
20 mM histidine/histidine-HCl, pH 5.9	PRO-0188988	8005030/ A-005478	25	Concentration and Diafiltration	Equilibration
					Diafiltration
					Recovery Flush
					Dilution to PFI
20 mM histidine/histidine-HCl, 960 mM sucrose, pH 6.0	PRO-0188988	8005513/ A-005465	25	Formulation	EAB Spike

16 REFERENCE DATA TRENDS

Sample chromatograms representing the three chromatography steps used in the purification process are presented below. It is preferred to provide the first cycle coming out of storage and the largest run scale.

Table 16-1 Protein A Chromatogram

Resin:	MabSelect Sure
Column Dimensions:	10 cm x 19cm
Notes:	Sani and storage not shown

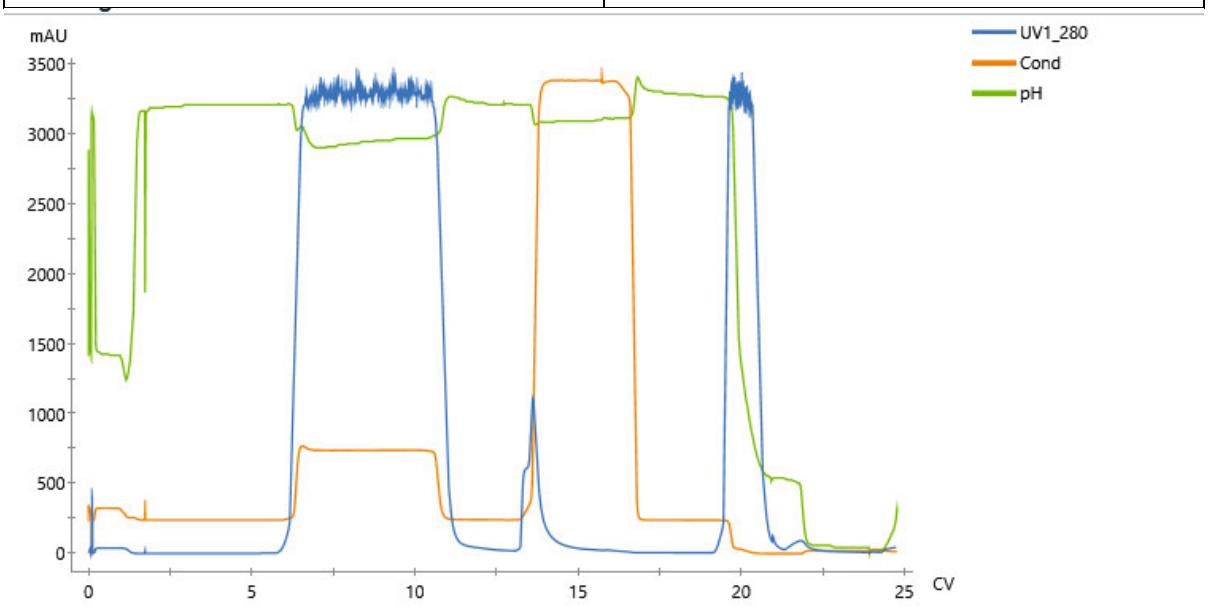


Figure 16-1 Mab Select Sure chromatogram

Low pH titration curves:

Scale: DSC 200L

Acid addition:

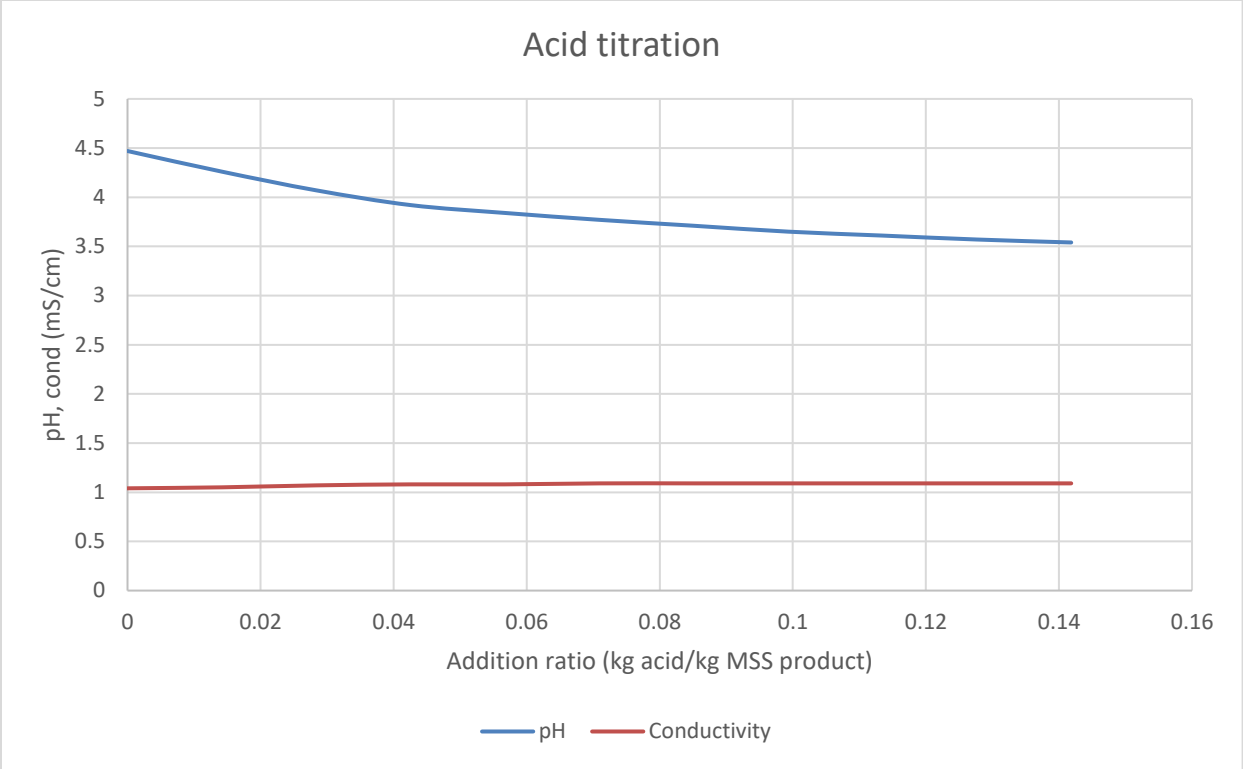


Figure 16-2 Low pH acid titration curve

Base Addition:

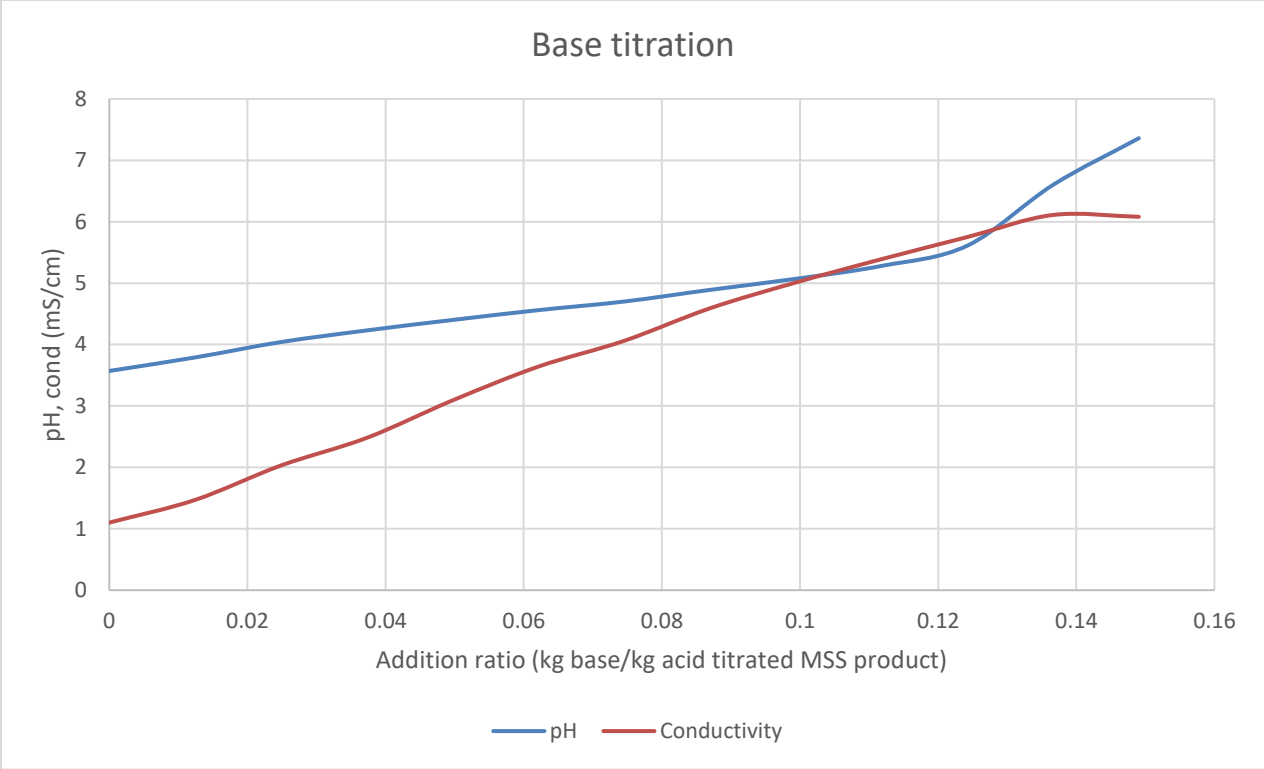


Figure 16-3 Low pH base titration curve

Table 16-2 Anion Exchange Chromatogram

Resin:	Capto Q
Column Dimensions:	0.66 cm x 19cm
Notes:	none

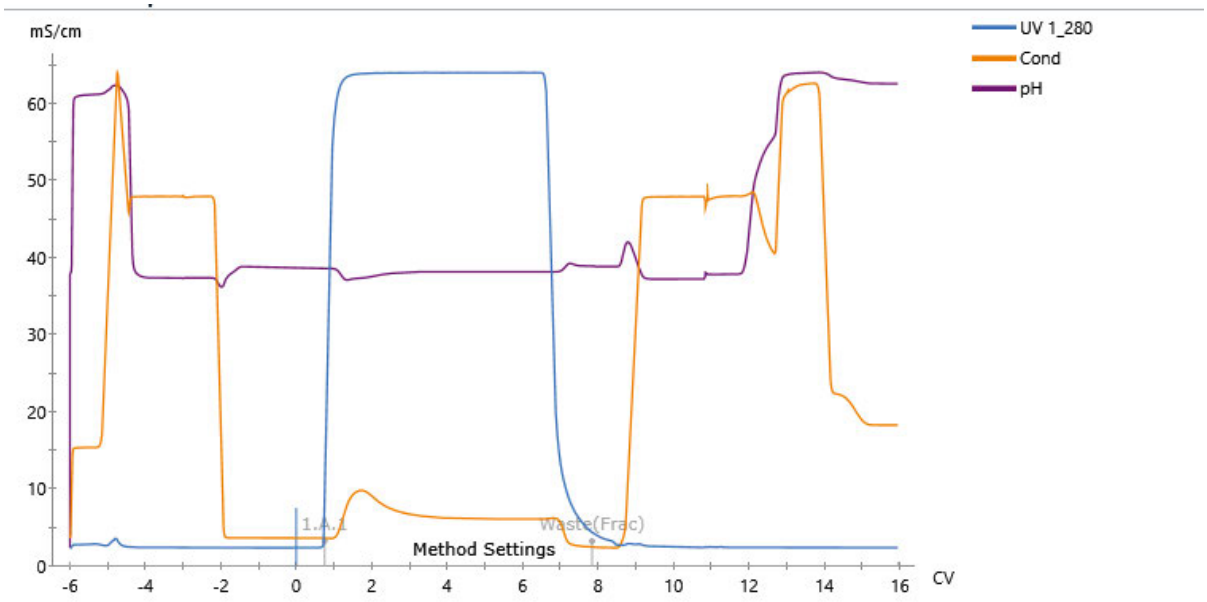


Figure 16-4 Capto Q Chromatogram

CEX Acid Titration: Scale: DSC 200L

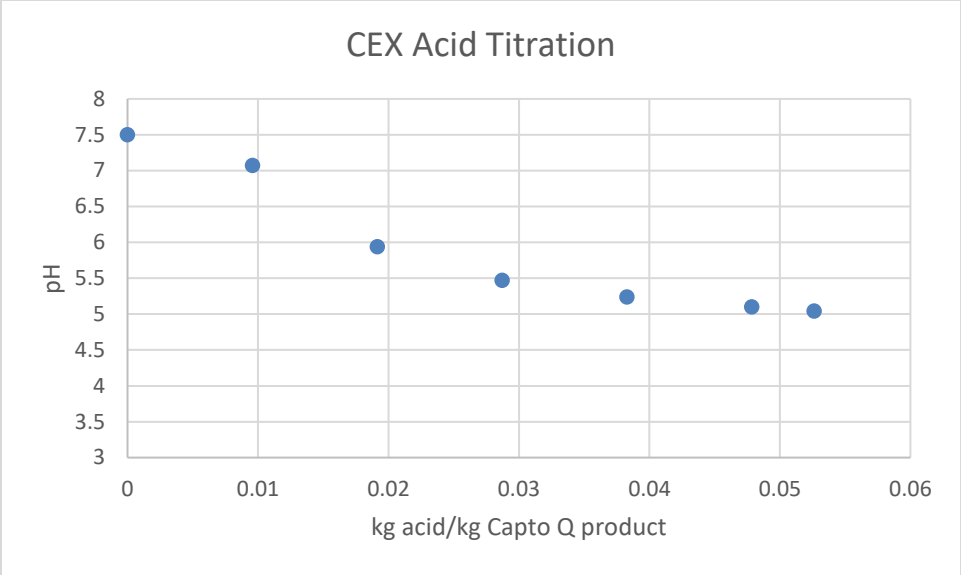
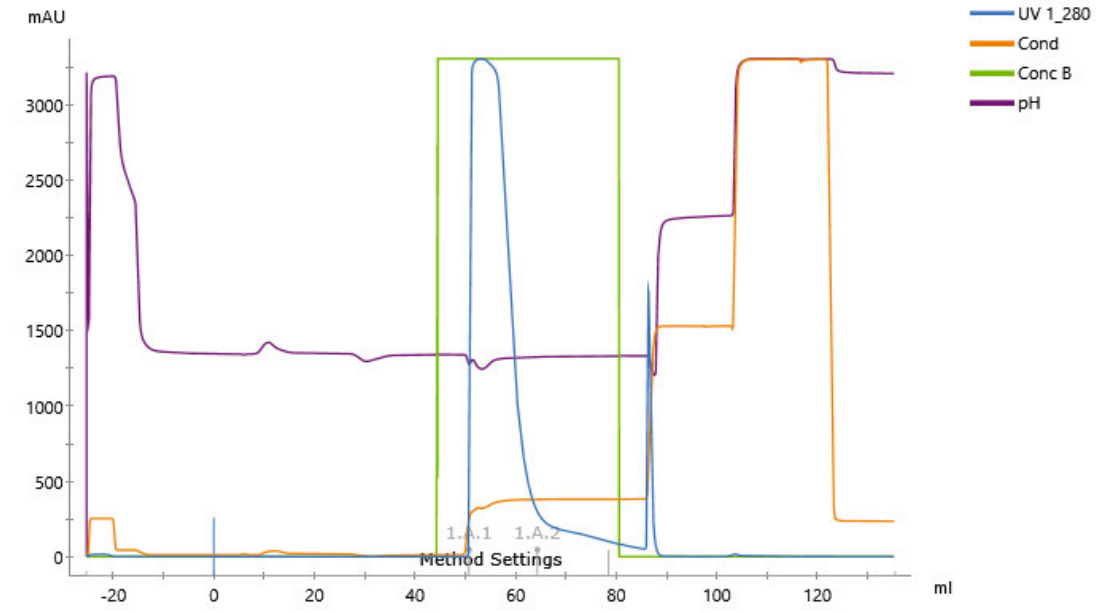


Figure 16-5 CEX load adjustment Curve

Table 16-3 Cation Exchange Chromatogram

Resin:	POROS 50HS
Column Dimensions:	0.66 x 19 cm
Notes:	none



17 PROCESS CHANGES

Table 17-1 lists changes in the INT-016 process from edition 000 to edition 6.0. These changes warrant a process number change from Process 0 to Process 2.

Table 17-1 Process Change Summaries and Justifications

Unit Operation	Step	Parameter	Original	New	Summary/Justification
AEX	Pre-Storage	Buffer	**ED 000**: None	**ED 001:** 50 mM tris, 0.5M sodium chloride, pH 7.4	Reduce quantity of storage buffer required
AEX	Storage	Buffer	**ED 000**: None	**ED 001: 100 mM sodium acetate, 2% (v/v) benzyl alcohol, pH 5.0**	Platform buffer to change soon
AEX	WFI Flush	Buffer	**ED 5.0 :** None	**ED 6.0 : WFI Flush**	Change made to comply with facility fit requirements of commercial site (FMC)
AEX	Pre-Equilibration	Buffer	**ED 5.0 :** **Target :** 5 CVs **NOR:** ≥ 4.5 Cvs	**ED 6.0** **Target :** 3.0 CVs **NOR:** ≥ 2.5 CVs	Change made to comply with facility fit requirements of commercial site (FMC)
CEX	Equilibration	Buffer	**ED 000:** 50 mM sodium acetate, pH 5.0	**ED 001:** 50 mM sodium acetate, 50 mM tris pH 5.0	Reduce/eliminate pH transient observed during elution
CEX	Elution	Buffer	**ED 000:** 50 mM sodium acetate, 200 mM sodium chloride, pH 5.0	**ED 001:** 50 mM sodium acetate 50 mM tris, 200 mM sodium chloride, pH 5.0	Reduce/eliminate pH transient observed during elution
CEX	Salt Strip	Buffer	**ED 000:** None	**ED 001:** 50 mM sodium acetate, 1 M sodium chloride, pH 5.0	Ensure proper stripping of column prior to sanitization
CEX	Product Adjustment	Buffer	**ED 000:** None	**ED 001:** 50 mM sodium acetate 50 mM tris, 200 mM sodium chloride, pH 5.0	Improve virus filter flux and reduce flux decay
VI	Incubation	Temperature	ED 001: NOR: 15-25 C	ED 002: NOR: 18-25 C	Recommendation based on recent projects

Unit Operation	Step	Parameter	Original	New	Summary/Justification
Protein A	Filtration	Load Filter	ED 4.0 : SHC	ED 5.0 : SHC + X0HC depth filter	Change made to comply with facility fit requirements of commercial site (FMC)
Protein A	Filtration	Product Filter	ED 4.0 : X0HC + SHC	ED 5.0 : SHC	Change made to comply with facility fit requirements of commercial site (FMC)
AEX	Pre-Storage	Buffer	ED 4.0 : 50 mM tris, 0.5 M Sodium Chloride, pH 7.4	ED 5.0 : WFI or 50 mM tris, 0.5 M Sodium Chloride, pH 7.4	Platform definition includes WFI as an option for Capto Q Pre-Storage.
UFDF	Membrane	Capacity	ED 4.0 : NOR max: 850 L/ m^2	ED 5.0 : NOR max 1600 L/m^2	Updated to tested capacity to fit requirements for commercial site (FMC)
BULK	General	mAb intermediate container	ED 5.0 : HDPE bottles	ED 6.0 : CryoVaults	Updated to the Process 2 container closure.
BULK	Buffer Flush	Flush	ED 5.0 : None	ED 6.0 : 20 mM histidine/histidine-HCl, 960 mM sucrose, pH 6.0	Change made to comply with facility fit requirements of commercial site (FMC)
BULK	Product flush	Flush	ED 5.0 : None	ED 6.0 : Product flush	Change made to comply with facility fit requirements of commercial site (FMC)

18 VERSION HISTORY

Version	Reason
1.0	• New document
2.0	• Updated Viral Inactivation temperature and duration parameters. For temperature the lower NOR limit changed from 15 to 18. For duration, the target was updated from 45 minutes to 70, AL range from 30-60 to 65-120 and NOR from 15-240 to 60-240.
3.0	• Virus filtration volumetric load AL added Language corrected from “drug substance” “unformulated drug substance” “DS” or “UDS” to “mAb intermediate” throughout document Added amount by volume of 0.67 mL/kg of acetic acid, glacial for buffer 50 mM Sodium Acetate, 50 mM Tris, 200 mM NaCl, pH 5.0 • Corrected Protein A Wash Buffer name to 50 mM tris, 500 mM sodium chloride, pH 7.4 • Removed CEX and Salt Srip from Table 15-1 Unit Operation and Steps section of 50 mM tris, 0.5 M sodium chloride, pH 7.4 • Updated expected titer range • Updated Conditioned Media Room Temperature Hold Time to 1 day in Table 3-1 • Updated 0.1 N sodium hydroxide Room Temperature Hold Time to 7 days (stedim bag) and 90 days (hyclone bag) in Table 15-1 • Updated MEDI PN/Document Number to 8008534/ AZDoc0158942 and Room Temperature Hold Time to 45 days for 50 mM sodium acetate, 50 mM Tris, pH 5.0 in Table 15-1 and Table 15-2 • Updated MEDI PN/Document Number to 8008535/ AZDoc0158136 and Room Temperature Hold Time to 45 days for 50 mM sodium acetate, 50 mM Tris, 200 mM sodium chloride, pH 5.0 in Table 15-1 and Table 15-3 • Updated Room Temperature Hold Time to 25 days for 20 mM histidine/histidine-HCl, pH 5.9 in Table 15-1 • Updated MEDI PN to GPF SPR PN 8005513 and Room Temperature Hold Time to 25 days for 20 mM histidine/histidine-HCl, 960 mM sucrose, pH 6.0 in Table 15-1
4.0	• Updated AEX pre-Storage column volumes from 2 to 3 CVs and adjusted min and max AL and NOR accordingly. • Updated AEX storage column volumes from 3 to 5 CVs and adjusted min and max AL and NOR accordingly per DEV000_GB_0469 • Changed the INT-016 mAb intermediate density from theoretical 1.039 g/mL to experimental 1.046 g/mL per EHIVE03-294243812-2096 ED 001 • Offline review completed by Darin El-Agib 08/24/22
5.0	• Updated table 4-1 Filtrations parameters : 1. Updated the NOR-Max values for 0.2 um filtration for Protein A, Virus filtration, AEX, CEX, and VF load and product filters for non fouling streams to 3000 L/ m2 based on new strategy to allow fit at common manufacturing facilities. 2. Added max challenges at scale to the justification section for all filtration steps. 3. Removed depth filtration from Protein A product filtration. 4. Added X0HC depth filter as load filter for Protein A. 5. Updated Nor-Max value for X0HC depth filtration to TBD and added max challenge in justification notes. • Updated table 12-1 1. NOR max for UFDF challenge to 1600L/ m^2, tested capacity of the filters.

Version	Reason
	2. Updated AL max for UFDF challenge to 1400 L/ m^2 based on NOR increase. • Removed tables 15-2, 15-3, 15-4 buffer tables and updated table 15-1 with PRO-0188988 for buffer references. • Added WFI as an option for AEX pre-storage buffer in Table 7-1. • Updated Table 1-1 with updated titer information.
6.0	• Process number updated form process 1 to process 2 in step 1.1 and section 16. • Alert limit definitions and values removed throughout the process description following the new platform template. • Updated table 1-1 for the mAb intermediate presentation , from stored at ≤ -70 °C in HDPE bottles to ≤ -40 °C in CryoVaults. • Updated Table 4-1 Filtration parameters: Protein A chromatography Load filter 1 Volumetric Load Nor-Max updated from TBD to 558 L/ m^2 Virus Inactivation Product filter 1 Volumetric load Nor-Max updated from TBD to 320 L/m^2. Concentration and Diafiltration Product filter Volumetric load Nor-Max updated from TBD to 483 L/m^2. mAb Intermediate Formulation and Bulk Filtration Bulk Filtration Volumetric load Nor-Max updated from TBD to 641 L/m^2. Added annotation a and b to specify options based on facility fit. • Table 7-1: updated Pre-equilibration CVs from 5 CVs target to 3 CVs. Updated the NOR from 4.5 CVs to 2.5 CVs. Added WFI flush step post NaOH strip. • Table 9-1: added note for linear velocity target for Column bed height packed between 22-25 cm to mitigate pressure issues. • Table 14-1 , Bulk filtration: added buffer and product flush Updated the container closure from HDPE bottles to CryoVault • Updated Table 3-1 to include additional hold time points for Virus inactivation product pre filter, UFDF recovery product and formulated bulk product. • Table 13-1 : Added a NOR for formulated mAb intermediate concentration and updated the process output note.
7.0	• Table 3-1, Added language under justifications/notes for protein A product. • Table 3-1 reworded language for formulated product for clarification. • Table 14-1, Updated Drug substance storage temperature NOR from -50 to -30 °C to ≤ -35 °C per SPEC-0153869. Two storage conditions were provided to allow storage at -40°C and -80°C. Subsequently, updated Table 1-1 mAb intermediate presentation to match section 14. • Table 14-1, Updated the 2-8 °C storage prior to freeze from 14 days to 1 month based on formulation sciences stability data.

Document Approvals

Business Approval	[redacted]
Business Approval	[redacted]000
Business Approval	[redacted]
Business Approval	[redacted]0